

ABSTRACT

This report presents the design and implementation of a desktop-based Library Management System developed using Java Swing, JDBC, and MySQL. The primary objective of the system is to provide an efficient and user-friendly interface for managing library records, reducing manual effort and improving data organization.

The application supports essential operations such as adding, viewing, and deleting book records through a graphical user interface. The system is structured using a layered architecture, comprising Data Transfer Object (DTO), Data Access Object (DAO), Service, and User Interface (UI) layers. This separation of concerns ensures modularity, ease of maintenance, and scalability of the application.

The DAO layer is responsible for handling database interactions using JDBC, while the service layer incorporates business logic and input validation to maintain data integrity. The UI layer, built using Java Swing, enables users to interact with the system in an intuitive manner.

This project demonstrates the integration of a Java-based GUI with a relational database system and highlights the importance of structured software design in developing reliable and maintainable applications.